Monsoon forcing, depth-aware calibration and decision value in national-scale groundwater forecasting for India

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Abstract. Machine-learning forecasts of groundwater levels are increasingly proposed for water management in India, but three questions that decide their usefulness remain open: whether monsoon rainfall adds skill beyond the memory of the aquifer, whether forecast uncertainty can be trusted for shallow and deep wells alike, and whether the forecasts are worth acting on. We address all three with 414,106 quality-controlled readings from 6,438 Central Ground Water Board monitoring wells (1994–2025), a time-blocked design (train $\leq$ 2015, calibrate 2016–2018, test 2019–2025) and nested gradient-boosting models fed with gridded IMD rainfall and temperature and ENSO and Indian Ocean Dipole indices. A memory-only global model removes 0.54, 0.55 and 0.20 of the squared error of persistence at 3, 6 and 12 months. Observed monsoon forcing adds a small but robust gain, largest at 6 months (5.0 % lower squared error, better in 22 of 24 test campaigns), that holds under an alternative split and in unseen regions; a perfect rainfall forecast would add a further 5.5–9.6 %. The single-day rainfall field distributed with the widely used public compilation of these data carries no information (median 3.6 mm per year), which explains an earlier null result. A single conformal interval covers 88 % of readings in wells shallower than 5 m but only 46 % in wells deeper than 40 m; intervals pooled by depth and region and updated online with adaptive conformal inference keep every depth and region class within 1.6, 1.7 and 3.02 percentage points of the nominal 80 and 90 % levels. Static calibration drifted because the wet 2019–2022 monsoons followed a dry calibration period. Exceedance warnings carry substantial economic value once their probabilities are recalibrated on a rolling window (peak relative value 0.43 for deeper-than-usual levels at 3 months and 0.69 for crossing an 8 m pump-suction limit at 6 months), whereas raw probabilities lose value for most users. We recommend issuing warnings when the forecast probability exceeds the user's cost–loss ratio and release the code, forecasts and an interactive dashboard.

1 Introduction

has been documented from wells and from satellite gravimetry alike (Rodell et al., 2009; Asoka et al., 2017). Most of the country's aquifers are recharged during a single monsoon season, so the question a farmer, a district planner or a scheme such as the Atal Bhujal Yojana asks before the dry season is concrete: how deep will the water be after the next monsoon, how sure can we be, and is it worth acting on a warning?

Machine-learning models of groundwater levels have multiplied: a recent review synthesises 223 studies and finds feature selection and tuning underdeveloped (Jesse et al., 2025), and many studies use a few dozen wells, random or single splits and no

naive baseline. At national scale, Malakar et al. (2021) trained recurrent networks on more than 5 000 Indian wells, and a 2026 open benchmark compared tree ensembles, recurrent networks and the Chronos foundation model on the public compilation of Central Ground Water Board (CGWB) levels used here (Alphoenixx, 2026). Work on German networks has shown how global models with static site descriptors behave (Heudorfer et al., 2024; Gomez et al., 2024; Ohmer et al., 2026), and a recent study of Korean wells found that known future weather, not site-specific learning, is the main source of short-horizon skill (Kim et al., 2026).

Three gaps remain. First, it is not established how much monsoon forcing adds to groundwater memory at seasonal lead times across India; the national benchmark found that climate features added almost nothing, but we show that the rainfall field it relied on is a single-day snapshot. Second, interval forecasts are rarely checked for coverage conditional on aquifer depth, region or season, and none of the studies we found used distribution-free (conformal) intervals for groundwater levels. Third, no groundwater forecasting study we found has measured the decision value of its warnings, although the cost–loss framework has been standard in weather forecasting for decades (Richardson, 2000).

We therefore ask: (RQ1) How much skill does monsoon forcing add to groundwater memory, and at which horizons, depths, regions and seasons? (RQ2) Can prediction intervals hold their stated coverage for every depth and region class, and keep holding when the climate regime shifts? (RQ3) Are exceedance warnings worth acting on, for whom and at what trigger?

2 Data

2.1 Groundwater levels

We use the public compilation of CGWB groundwater levels (Alphoenixx, 2026) (Table 1). The file mixes two networks: manual wells read in four campaigns a year (January, May, August and November) and Digital Water Level Recorder (DWLR) telemetry wells read several times a day since about 2023. The two use opposite sign conventions, which we harmonise to depth below ground level by each station's majority sign. Physically impossible values (< -5 m or > 250 m) and per-station outliers (robust $z > 8$) are removed. Manual readings are mapped to the nearest campaign, giving 246,887 campaign values at 5,516 wells (Fig. 1). Forecasts are made for these manual wells; DWLR wells are used only for description.

2.2 Forcing

Daily rainfall on the IMD 0.25° grid (Pai et al., 2014) and daily maximum and minimum temperature on the 1° grid (Srivastava et al., 2009) were aggregated to monthly values for 1990–2025. Each well takes the nearest grid cell with data. In the rainfall grid, 362 cell-years recorded zero rain all year while their neighbours received hundreds of millimetres; these were replaced by the neighbours' mean, and 477 cell-years in contiguous zero blocks were set to missing. Monthly Niño 3.4 and Dipole Mode Index (DMI) series from HadISST (Rayner et al., 2003; Saji et al., 1999) represent ENSO and the Indian Ocean Dipole.

Three checks support the forcing (Fig. 2). All-India June–September departures from our grid agree in sign with IMD's published departures in all ten years checked. The May-to-November rise of the water table increases with the season's rainfall

Table 1. Data used in this study.

Item	Value
Stations (manual / DWLR)	6,438 (5,516 / 922)
Readings after quality control	414,106
Period	1994–2025
Manual-well campaign values (Jan, May, Aug, Nov)	246,887 from 5,516 wells
Split by target year	train $\leq$ 2015; calibration 2016–2018; test 2019–2025
Rainfall forcing	IMD 0.25° daily gridded, 1990–2025
Temperature forcing	IMD 1° daily Tmax/Tmin, 1990–2025
Teleconnections	Niño 3.4 and DMI (HadISST, NOAA PSL), lags 0/3/6 months

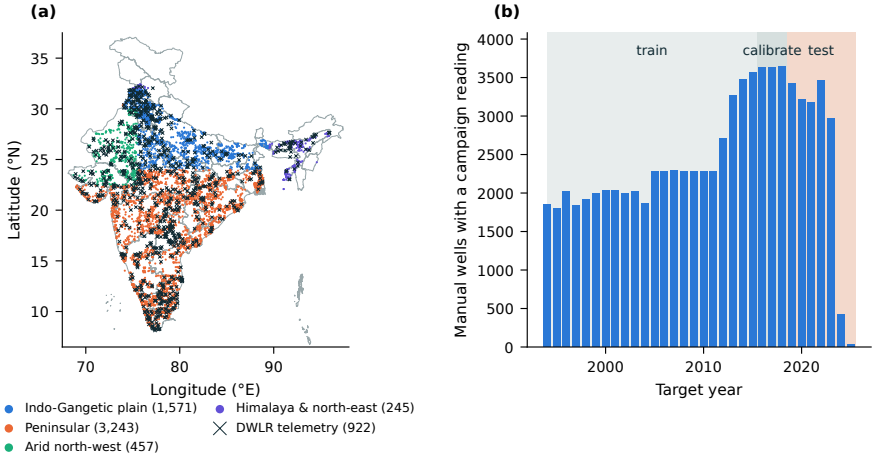


Figure 1. (a) Monitoring wells by region (manual network) and DWLR telemetry wells. (b) Manual wells with a campaign reading per target year and the time-blocked split.

at the well, positively at 78 % of wells. And the rainfall column distributed with the groundwater file is unrelated to real rainfall: summed over four campaigns it has a median of 3.6 mm, against 1,004 mm of IMD rain over the previous year. That column holds rainfall on the day of the reading.

3 Methods

3.1 Forecasting task and validation design

For each well and origin campaign t we forecast the depth at $t + h$ for $h = 1$, 2 and 4 campaigns (about 3, 6 and 12 months). Rows are assigned by the year of the target: training ≤ 2015, calibration 2016–2018 and test 2019–2025, the same boundary

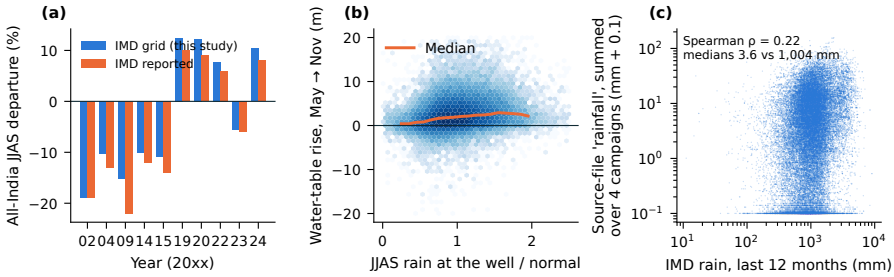


Figure 2. Forcing checks. (a) All-India JJAS rainfall departures from the IMD grid and as reported by IMD. (b) Monsoon recharge (May to November rise) against the season's rainfall relative to normal; line: binned median. (c) The rainfall column of the source file against IMD rainfall over the previous 12 months.

for every well. Features available at the origin are the current depth, four lags, changes over one and four campaigns, origin and target season, an expanding per-well seasonal climatology (no look-ahead), the current anomaly, record length, location and borehole depth. Forcing features use only months before the origin campaign month; climatologies and SPI parameters (McKee et al., 1993) are fitted on years up to 2015. Unit tests check that none of these features can see data after the origin.

3.2 Nested models (RQ1)

All models are global XGBoost regressors (Chen and Guestrin, 2016) that predict the change $y_{t+h} - y_t$. Five arms are trained on identical rows: *M0* memory only; *M1* adds past rainfall (since the previous campaign, 3, 6 and 12 months), the last monsoon total and its ratio to normal, SPI-3/6/12 and the temperature anomaly; *M2* adds Niño 3.4 and DMI at lags of 0, 3 and 6 months; *M3* adds the rainfall that actually fell between origin and target, an oracle that bounds the value of a perfect seasonal rainfall forecast; and the control *C* adds the source file's single-day rainfall summed over 2, 4 and 8 campaigns, as in earlier work. Each arm receives the same search over six settings, early-stopped on target years 2013–2015 and refitted on all training years. We compare against persistence, seasonal naive, per-well climatology, anomaly persistence, Random Forest and zero-shot Chronos-2 (Ansari et al., 2024).

Skill is the reduction in mean squared error relative to persistence. Confidence intervals come from a well-clustered bootstrap (1 000 draws). About 0.1 % of test readings are single-reading jumps of tens of metres, probably datum changes or replaced wells, which carry about 15 % of the squared error. They remain in every headline number, but they make squared-error Diebold–Mariano tests (Diebold and Mariano, 1995) uninformative, so we report the mean-absolute-error reduction with clustered intervals and a two-sided sign test on the number of test campaigns in which the richer arm has the lower error. Robustness is checked with an alternative split (train $\leq$ 2017, test 2020–2025) and with five-fold cross-validation over $2^\circ \times 2^\circ$ spatial blocks, which forecasts regions the model has not seen.

3.3 Interval calibration (RQ2)

Intervals wrap the M2 point forecast. Split conformal intervals (Lei et al., 2018) use one pool of signed calibration residuals with finite-sample ranks; Mondrian variants keep separate pools for depth class × season or depth class × region, with fallback to the parent pool below 50 residuals; we also test Gaussian intervals, residuals normalised by a cross-fitted spread model, and conformalised quantile regression (Romano et al., 2019). The online variant grows each pool with test residuals whose target lies at least h campaigns before the forecast's target (that is, already observed when the forecast was issued) and adapts each cell's working α with adaptive conformal inference (Gibbs and Candès, 2021), using a step size $\gamma = 0.05$ fixed before testing. Coverage is evaluated at 50, 80, 90 and 95 % overall and by depth, region, season and record length, with binomial tests and Holm correction, the Winkler score and CRPS (Gneiting and Raftery, 2007). Our acceptance criterion is that every depth and region class stays within 3 points of nominal at 80 and 90 %.

3.4 Warnings and decision value (RQ3)

We define three events at the target: E1, depth deeper than the well's own 80th percentile for that season; E2, a standardised groundwater index below -1 (Bloomfield and Marchant, 2013); and E3, for wells currently at 8 m or shallower, the water falling below 8 m, where centrifugal surface pumps lose suction. Thresholds use readings up to 2018. The raw probability of an event is the share of residuals in the online depth × region pool that would place the forecast beyond the threshold. Because interval calibration does not imply calibration at every threshold, we also recalibrate probabilities with isotonic regression (Zadrozny and Elkan, 2002), either once on 2016–2018 or on a rolling window of the eight most recent campaigns whose outcomes were known at issue time. Both recalibrations were added after the raw reliability diagram showed over-confidence; the window length was fixed before the rolling variant was run, and no test outcome enters a forecast before it was observable. Probabilities are scored with the Brier skill score (Brier, 1950) against each well's own climatological frequency and with ROC AUC. Decision value is the relative economic value V of Richardson (2000) for a user who pays C to protect against a loss L and acts when $p \geq C/L$; $V = 1$ is a perfect forecast and $V = 0$ is no better than always or never acting. Intervals come from a paired well-clustered bootstrap.

4 Results

4.1 How much does monsoon forcing add? (RQ1)

Groundwater memory carries most of the skill (Table 2, Fig. 3). The memory-only model removes 0.54, 0.55 and 0.20 of persistence's squared error at 3, 6 and 12 months, well ahead of anomaly persistence. Adding observed forcing and ENSO/IOD (M2) improves on memory alone in every horizon and setting (Table 3): by 1.1, 5.0 and 1.3 % of squared error and by 2.0, 3.6 and 1.5 % of absolute error, with absolute-error intervals that exclude zero at all horizons. The gain is clearest at 6 months, where the forcing arm is better in 22 of 24 test campaigns ($p < 0.001$) and remains positive under the alternative split (3.1 %) and in unseen regions (3.7 %). Rainfall alone (M1) adds less and loses its 12-month gain in unseen regions.

Table 2. Test-period (2019–2025) point skill on identical rows: RMSE (m) and MSE skill against persistence with 95 % well-clustered bootstrap intervals.

Model	RMSE 3 mo	Skill 3 mo	RMSE 6 mo	Skill 6 mo	RMSE 12 mo	Skill 12 mo
Persistence	3.48	0	4.17	0	3.17	0
Anomaly persistence	2.69	0.404 [0.354, 0.446]	3.11	0.444 [0.393, 0.484]	3.17	0.000 [0.000, 0.000]
C: snapshot rainfall	2.38	0.531 [0.496, 0.561]	2.80	0.550 [0.512, 0.584]	2.87	0.185 [0.143, 0.226]
M0: memory	2.37	0.536 [0.503, 0.563]	2.80	0.549 [0.506, 0.586]	2.84	0.201 [0.166, 0.235]
M1: + past forcing	2.36	0.539 [0.503, 0.568]	2.75	0.564 [0.526, 0.598]	2.82	0.212 [0.179, 0.244]
M2: + ENSO/IOD	2.36	0.541 [0.507, 0.569]	2.73	0.572 [0.529, 0.608]	2.82	0.212 [0.179, 0.243]
M3: + oracle rain	2.29	0.566 [0.536, 0.592]	2.65	0.598 [0.559, 0.633]	2.68	0.287 [0.256, 0.316]

Rows: 25,340 / 24,073 / 32,472 forecasts at 3 / 6 / 12 months.

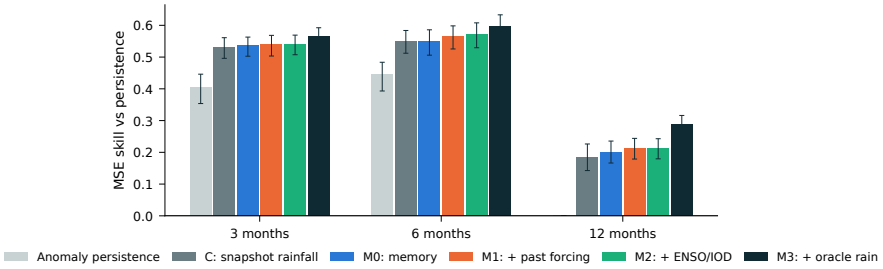


Figure 3. Skill against persistence by arm and horizon on identical test rows; bars are 95 % well-clustered bootstrap intervals.

The gain is physically placed (Fig. 4). It concentrates in the forecast that spans the monsoon, May to November, where M2 cuts squared error by 12 % at 6 months, and in wells shallower than 15 m. Deep wells and the Himalaya and north-east gain little. Knowing the rain that actually fell (M3) would add a further 5.5–9.6 % in the main split (at least 2.5 % in every robustness setting), growing with horizon; this is the skill a seasonal monsoon forecast could unlock. The single-day rainfall of the source file adds nothing (C vs M0: -2.1 to $+0.2$ %), which reproduces the earlier null result and shows that it came from the feature, not the hydrology. Chronos-2 used zero-shot trails the trained models (skill 0.40 at 3 months against 0.51 for XGBoost on the same rows), although rainfall covariates reduce its absolute error by 6–7 % at 3 and 6 months, and Random Forest performs like XGBoost.

4.2 Can the intervals be trusted for every well? (RQ2)

A single interval width fails by depth (Fig. 5, Table 4): the 80 % split-conformal interval covers 88 % of readings in wells shallower than 5 m but 46 % in wells deeper than 40 m, and Gaussian intervals over-cover by ten points overall because the datum jumps inflate the residual spread, yet still fail by depth. Pools by depth and region repair most of this, but static calibration under-covers by 2–5 points overall. The cause is a change of regime (Fig. 6): the calibration years 2016–2018 followed the

Table 3. Skill added between nested arms. MSE and MAE reduction (%, MAE with 95 % well-clustered interval), test campaigns in which the richer arm has lower MAE with a two-sided sign test, and MSE reduction under an alternative split and spatially blocked validation.

Comparison	Horizon	MSE	MAE	Campaigns	Sign p	Alt. split	Unseen regions
M1 vs M0	3 months	+0.7	+1.2 [+0.6, +1.8]	13/24	0.839	+2.8	+0.2
	6 months	+3.3	+1.5 [+1.0, +2.0]	19/24	0.007	+1.8	+2.2
	12 months	+1.3	+1.9 [+1.5, +2.2]	22/25	<0.001	+2.7	−1.4
M2 vs M0	3 months	+1.1	+2.0 [+1.4, +2.6]	16/24	0.152	+3.0	+2.0
	6 months	+5.0	+3.6 [+3.1, +4.0]	22/24	<0.001	+3.1	+3.7
	12 months	+1.3	+1.5 [+1.1, +2.0]	17/25	0.108	+5.7	+1.0
M3 vs M2	3 months	+5.5	+3.4 [+2.9, +3.9]	19/24	0.007	+2.8	+2.5
	6 months	+6.1	+4.2 [+3.5, +4.8]	15/24	0.307	+5.2	+5.9
	12 months	+9.6	+6.5 [+5.8, +7.1]	20/25	0.004	+9.3	+5.5
C vs M0	3 months	−1.0	−0.5 [−0.9, −0.1]	10/24	0.541	–	–
	6 months	+0.2	+0.5 [+0.0, +0.9]	14/24	0.541	–	–
	12 months	−2.1	+0.1 [−0.2, +0.5]	12/25	1.000	–	–

2014–2015 droughts, when water tables came in deeper than forecast, while the strong 2019, 2020 and 2022 monsoons left water shallower than forecast, by a median of up to 0.68 m at 12 months in 2022.

The online depth × region method with adaptive conformal inference absorbs the shift. It keeps every depth and region class within 1.6, 1.7 and 3.02 points of nominal at 80 and 90 % at 3, 6 and 12 months. The one cell beyond 3 points is the Himalaya and north-east at 12 months, which over-covers at 83.0 % with 854 readings (Holm-adjusted $p = 0.23$); we did not retune γ on the test period to remove it. Two variants added after inspecting the results (Table 4, marked) also fix residual season misses and sharpen intervals, at the cost of slightly larger region misses. Conformalised quantile regression and normalised conformal intervals are sharper but miss individual depth or region classes by up to 9.4 points.

4.3 Are the warnings worth acting on? (RQ3)

Raw warning probabilities are over-confident (Fig. 7): a predicted 55 % verified 33–41 % of the time. Recalibration once on 2016–2018 over-corrects on the test years, the same drift as in RQ2, whereas the rolling recalibration follows the diagonal and raises the Brier skill score at every horizon (Table 5). With rolling-recalibrated probabilities, warnings carry positive value for nearly every cost–loss ratio (Fig. 8). Peak relative value is 0.43 for E1 at 3 months and 0.69 for E3 at 6 months. Observed forcing adds a small value gain at 6 months only (+0.01 to +0.04 for C/L between 0.1 and 0.5), and the oracle rainfall arm adds up to 0.13, again pointing to seasonal rainfall forecasts as the next source of value.

At a trigger of 0.3, three-month E1 warnings are issued for about a quarter of forecasts, catch 58 % of events with 46 % of warnings not verifying, and give $V = 0.37$. At a trigger of 0.2, six-month E3 warnings catch 76 % of crossings of the suction limit with 48 % of warnings not verifying ($V = 0.58$). In the dry 2023–2024 water year, exceedances in the Atal Bhujal states

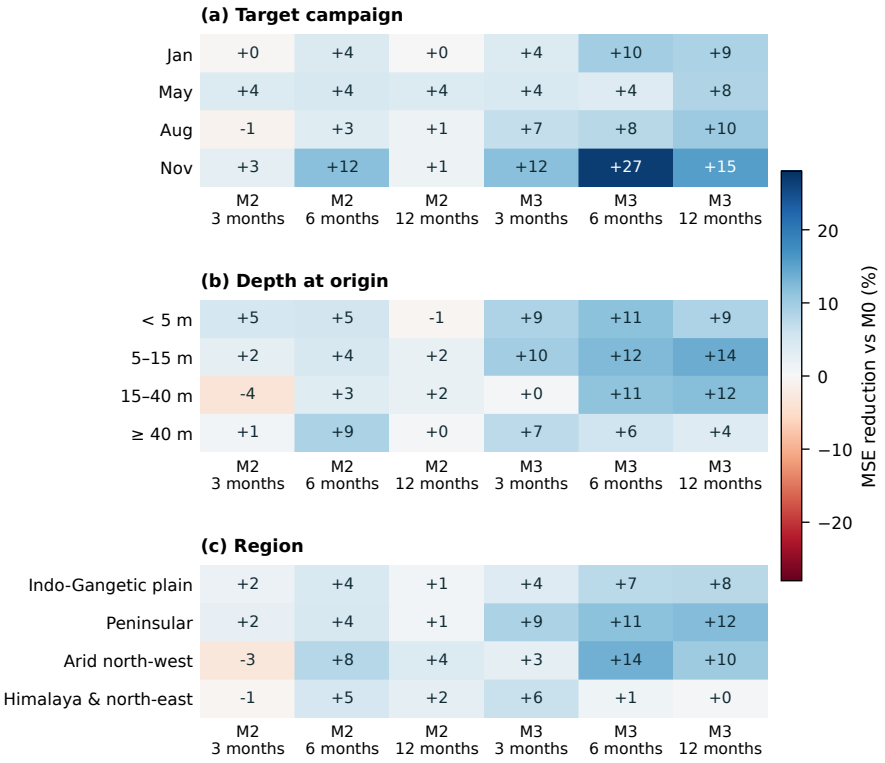


Figure 4. Reduction in squared error over the memory-only model by (a) target campaign, (b) depth at the origin and (c) region, for M2 (observed forcing) and M3 (oracle rainfall).

rose from 17 % to 25 % of readings while the mean warning probability barely moved (0.16 to 0.17): the forecasts lagged the turn to a dry regime.

5 Discussion

5.1 Memory first, forcing second

At national scale and seasonal lead times, the state of the aquifer explains most of what can be forecast. Observed rainfall adds a few per cent, concentrated where hydrology says it should: shallow wells and the monsoon-spanning forecast. That is consistent with Korean wells (Kim et al., 2026), where past weather added little and known future weather added most, and it reframes the value of climate information for Indian groundwater: the gain lies in forecasting the coming monsoon, not in describing the last one.

Table 4. Interval calibration in the test period: overall coverage of the 80 % interval (%), the worst absolute deviation from nominal over depth and region classes at 80 % and 90 % (points), and the Winkler score (m).

Method	Cov.	Worst	Winkler	Cov.	Worst	Winkler	Cov.	Worst	Winkler
Gaussian	90.0	26.2	9.20	89.5	19.3	10.23	89.2	21.4	10.22
Split conformal	80.2	33.6	8.62	78.3	28.6	9.79	77.5	30.9	10.00
Mondrian D×S	78.0	10.9	8.12	76.7	11.4	9.29	75.4	15.1	9.44
Mondrian D×R	78.0	8.8	8.05	76.5	4.9	9.18	75.2	5.8	9.34
Normalised	78.0	6.2	7.29	77.2	7.1	8.24	76.6	8.8	8.25
CQR	80.5	4.4	7.03	79.5	6.2	7.99	80.3	9.4	8.02
Mondrian D×R + recal	78.9	2.3	7.97	77.9	2.9	9.02	77.0	4.0	9.11
Mondrian D×R + recal + ACI	79.3	1.6	7.99	79.9	1.7	9.04	78.7	3.0	9.12
Mondrian D×R×S + recal + ACI†	78.6	2.6	8.00	77.9	2.9	9.08	77.5	3.6	9.05
Normalised D×R×S + recal + ACI†	78.4	2.1	7.29	78.0	3.6	8.18	78.4	3.3	8.02

Column groups: 3, 6 and 12 months. † Added after inspecting results (post hoc).

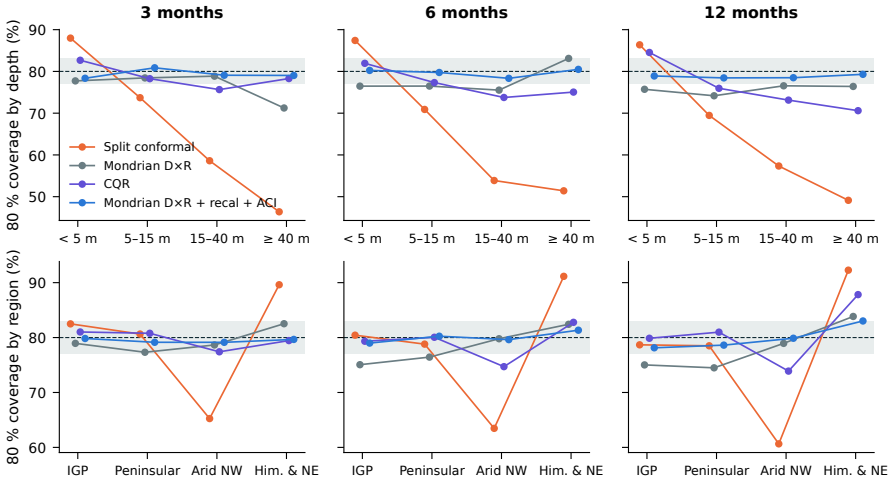


Figure 5. Coverage of 80 % intervals by depth class (top) and region (bottom); the grey band marks ±3 points around the target.

5.2 Reproducing a null result

The 2026 benchmark on the same file found that climate features added almost nothing. We reproduce that result with the same construction and show that the feature is a single-day reading. Benchmarks built on public compilations need a check of what each column measures.

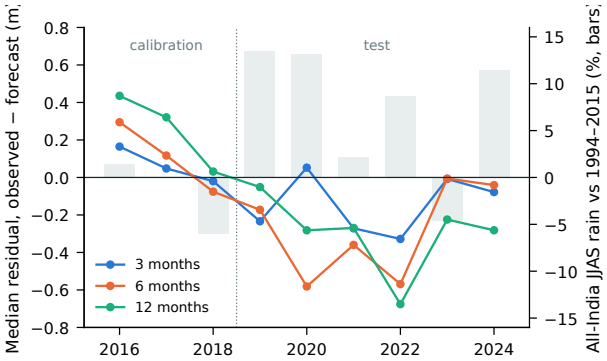


Figure 6. Median residual (observed minus forecast depth) of M2 by target year, with all-India JJAS rainfall anomaly relative to 1994–2015 as bars.

Table 5. Warning skill and value for model M2 with rolling-isotonic probabilities: event base rate, Brier skill score against each well's climatology (95 % interval), ROC AUC, peak relative economic value and the cost–loss ratio at which it occurs, and the range of cost–loss ratios with positive value.

Event	Horizon	Base rate	BSS	AUC	$V_{\max}$	at C/L	$V > 0$ for C/L
Deeper than usual	3 months	0.24	0.234 [0.216, 0.250]	0.79	0.43	0.24	0.02–0.96
	6 months	0.24	0.165 [0.149, 0.183]	0.74	0.32	0.24	0.06–0.92
	12 months	0.25	0.143 [0.126, 0.159]	0.71	0.27	0.24	0.06–0.98
SGI < -1	3 months	0.19	0.265 [0.239, 0.293]	0.81	0.45	0.18	0.02–0.92
	6 months	0.19	0.207 [0.181, 0.233]	0.78	0.38	0.18	0.04–0.90
	12 months	0.19	0.185 [0.158, 0.212]	0.75	0.32	0.20	0.04–0.90
Below 8 m	3 months	0.09	0.336 [0.306, 0.365]	0.91	0.65	0.08	0.02–0.68
	6 months	0.10	0.255 [0.224, 0.287]	0.92	0.69	0.10	0.02–0.74
	12 months	0.06	0.253 [0.220, 0.285]	0.86	0.54	0.06	0.02–0.46

5.3 Calibration under regime change

Conformal guarantees assume that calibration and test residuals are exchangeable. Monsoon variability breaks that assumption at the scale of years, and the break is visible in the residuals. Online recalibration with adaptive conformal inference is cheap and restored conditional coverage without re-training; the same mechanism applied to probabilities restored decision value. Interval coverage alone is not enough evidence for warning probabilities.

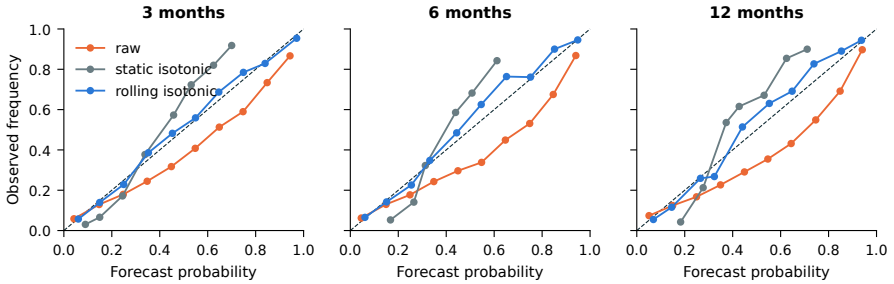


Figure 7. Reliability of E1 warning probabilities from M2: raw, recalibrated once on 2016–2018, and recalibrated on a rolling window.

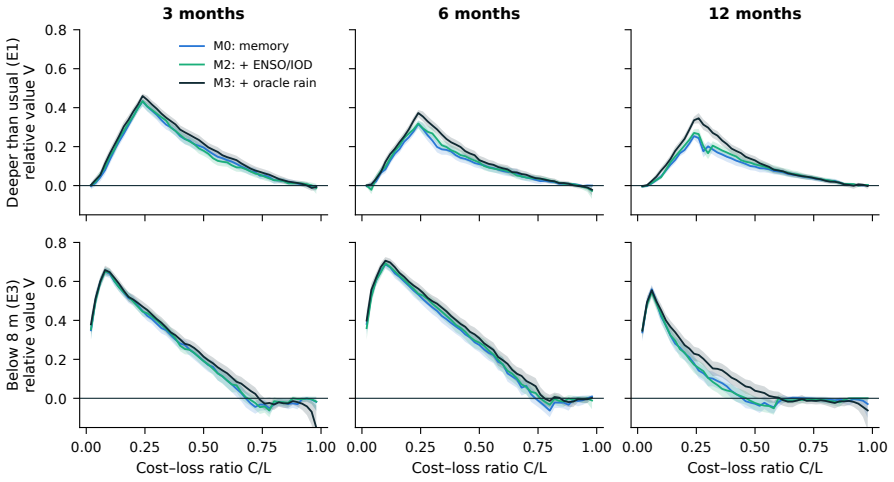


Figure 8. Relative economic value of E1 and E3 warnings against the user's cost–loss ratio, for the memory-only, forcing and oracle-rainfall models; bands are 95 % well-clustered intervals.

5.4 Limitations

Abstraction is not observed, so the models cannot separate a pumping response from a climate response and cannot support pumping-quota optimisation. Most manual records in the compilation end in 2022–2023, so test campaigns after 2023 are thin. Regions are coarse latitude–longitude boxes, not aquifer polygons. The datum jumps were kept rather than cleaned. Telemetry wells were not used for forecasting. The case study is at state level because an official district list of the Atal Bhujal scheme was not available to us.

6 Conclusions

(1) Groundwater memory explains most seasonal forecast skill across India; observed monsoon forcing adds a small, robust gain, largest at 6 months, and a perfect monsoon forecast would add 5.5–9.6 % more. (2) The rainfall field of the public compilation is a single-day snapshot and should not be used as forcing. (3) Conformal intervals pooled by depth and region and updated online keep every depth and region class within about 3 points of nominal. (4) Warnings are worth acting on once their probabilities are recalibrated on recent outcomes; the rule "warn when $p \geq C/L$" serves every user and needs no tuning; triggers of 0.3 for seasonal planning and 0.2 for pump decisions six months ahead are illustrative, read from the test period.

Code and data availability. Code, processed results, figures and the dashboard are in the project repository (archive and DOI to be added on release). The groundwater levels come from a public compilation of CGWB data; IMD gridded data were obtained with `imdlib`; Niño 3.4 and DMI from NOAA PSL; state boundaries from geoBoundaries (Runfola et al., 2020). Redistribution terms of the groundwater compilation are being confirmed before processed data are archived.

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